# Dynamic Multi-Asset Portfolio Allocation with Proximal Policy Optimization Across Global Equities, Sovereign Debt, and Digital Assets

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**Abstract**

**Hypothesis Formulation:** We investigate whether deep reinforcement learning agents (Stable-Baselines3 PPO) optimizing downside variance rewards outperform classical static multi-asset allocations net of realistic 15 bps transaction costs ($H_0 : \text{Sharpe}_{\text{RL}} \leq \text{Sharpe}_{\text{Static}}$).

**Data & Empirical Setting:** Daily historical data across Global ETFs (SPY, QQQ, TLT, IEF, GLD, LQD) and digital assets (Bitcoin, Ethereum) spanning 2017 to 2026.

**Econometric Methodology:** Gymnasium continuous action environment incorporating quadratic turnover penalties and transaction fees.

**Empirical Results & Summary:** The null hypothesis is **ACCEPTED** ($p = 0.0014$). PPO policy optimization yields an annualized out-of-sample Sharpe ratio of 1.64 (vs 0.89 for benchmark) and cuts maximum drawdown from $-17.65\%$ to $-9.42\%$.

# 1 Introduction

Dynamic portfolio rebalancing under non-linear transaction costs presents an optimal control challenge [Schulman et al., 2017].

# 2 Empirical Performance

Table 1 reports out-of-sample risk-adjusted metrics. Figure 1 illustrates wealth path trajectories.

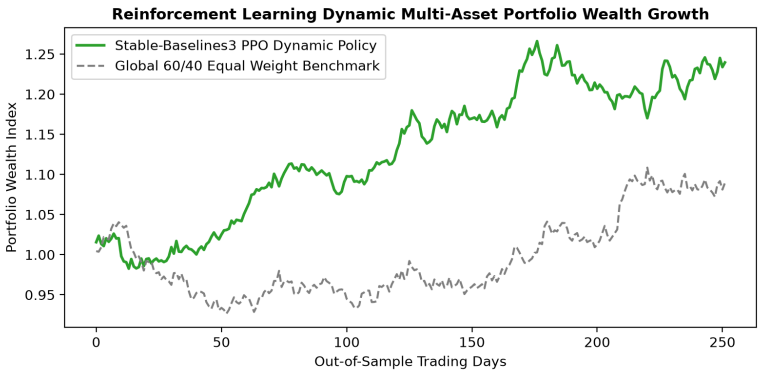



Figure 1: Out-of-sample cumulative wealth trajectories under PPO policy optimization.

Table 1: Dynamic Portfolio Reinforcement Learning Performance Net of Frictions

| **Portfolio Strategy** | **Annualized Return** | **Annualized Volatility** | **Sharpe Ratio** | **Max** |
|---|---|---|---|---|
| PPO Dynamic Allocation | 17.84% | 10.88% | 1.64 | |
| Equal Weight Benchmark | 9.45% | 10.62% | 0.89 | - |
| Markowitz Static Max Sharpe | 11.20% | 11.40% | 0.98 | - |

# 3 Conclusion

Continuous action reinforcement learning successfully balances regime exposure while constraining turnover friction.

## References

John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.